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Hazard Pointers for C++26

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1 Introduction

We propose the hazard pointer safe deferred reclamation technique [1] for inclusion in C++26. This paper contains proposed interface and wording for hazard pointers and rationale for its inclusion in C++26. The wording is based on N4700 ("Working Draft, Standard for Programming Language C++"). An implementation is in the Facebook Folly open source library [2] and has been in heavy use in production since 2017.

The proposal is a subset of the hazard pointer interface and wording in N4895 ("Working Draft, Extensions to C++ for Concurrency Version 2") (henceforth referred to as TS2), which is based on P1121R3 ("Hazard Pointers: Proposed Interface and Wording for Concurrency TS 2"). The proposed interface and wording are in Section 4.

1.1 Feature-Test Macro

We propose a new feature-test macro `__cpp_lib_hazard_pointer` be added to Section 17.3.2 of the IS.

1.2 Change History

Changes from P2530R2 to P2530R3:

- Change the title from "Why Hazard Pointers Should be in C++26" to "Hazard Pointers for C++26".
- Change to the wording based on feedback from LWG review meetings on 2023-02-07, 2023-02-08 (Issaquah) and 2023-03-01.

Changes from P2530R1 to P2530R2:

- Add a comparison table with and without hazard pointers to the use examples.

Changes from P2530R0 to P2530R1:

- Add a feature-test macro.
- Present the wording relative to the standard instead of being relative to N4895 (Concurrency TS 2 draft).
- Add a code example demonstrating user code iterators based on hazard pointers and provide latency estimates.
- Add guidance for future ABI stability.

1.3 Reference Implementation

The reference implementation of hazard pointer that includes a superset of the proposed interface is in the Folly open-source library (<https://github.com/facebook/folly/blob/main/folly/synchronization/Hazptr.h>).

1.4 Rationale for inclusion in C++26

This proposal is based on the experience gained from hazard pointer implementation in the Facebook Folly open-source library (since 2016) and its heavy use in production (since 2017). The Folly implementation includes the TS2 interface (which was functionally frozen in late 2017) and significant subsequent functional extensions. The implementation of the selected subset of TS2 proposed for C++26 has been stable and in heavy continuous use in production since 2017.

What are the reasons for selecting the proposed subset of the TS2 interface for C++26?

- The subset supports a large part of hazard pointer usage in production.
- The subset interface and implementation have been stable for years under heavy use in production.
- The changes to the existing TS2 wording are straightforward (as can be seen in Section 4 of this paper) and are therefore expected to require little of the committee's time.

Which parts of the TS2 interface are omitted from this proposal for C++26 and why?

- This proposal omits **custom domains**:

- The Folly implementation experience has been that we were able to improve and extend the default domain, without needing custom domains. This does not preclude the need for custom domains for some use cases (e.g., `async-signal-safety`).
- That is, we did not include custom domain for lack of evidence they are not necessary for common usage.
- This proposal does not preclude adding custom domains in the future, as evident in TS2.
- This proposal also omits **global cleanup**:
 - The TS2 interface includes a global cleanup function (`hazard_pointer_clean_up`) which guarantees that all reclaimable retired objects are reclaimed before the function returns. We refer to this mode of reclamation as synchronous reclamation in contrast to the default asynchronous reclamation, where retired objects are only guaranteed to be reclaimed by the end of the program.
 - In 2018 we introduced to Folly cohort-based synchronous reclamation. Cohort-based reclamation is faster and more scalable than global cleanup and can be used conservatively. It has been in heavy use in production since 2018.
 - We do not include cohort-based synchronous reclamation in this proposal for the following reasons:
 - * The wording is expected to be complex and would take a nontrivial amount of committee time.
 - * Use cases that do not require synchronous reclamation are important. There is no need to delay inclusion into the IS to support such cases.
 - * This proposal is amenable to adding future support for synchronous reclamation.

Why the rush to get it into C++26 when the Concurrency TS 2 is not published as yet (as of early 2022)?

- We were able to use the TS2 interface (functionally unchanged since 2017) and gain feedback for the Facebook Folly library implementation under heavy use in production.
- This production experience has enabled us to reduce the TS2 interface already to a stable subset that is currently heavily used in production.
- The plan was always to include a basic interface into C++ IS, and we have achieved that with this experience.
- With hazard pointers in C++26, it enables us to evolve it further and we have a growing list of features and improvements which would be hard to add without being in the IS.
- We can continue to incorporate feedback experience from other usage and evolve it.
- The straightforward changes to the interface surface area (all subtractions) means that the wording review effort is small, and can take little time to be reviewed for C++26 IS inclusion.
- The subset proposed for C++26 is amenable to potential future extensions based on feedback to TS2.
- Note that this selection does not constitute a rejection of the components of TS2 not selected for the C++26 proposal. Rather, we cannot recommend the excluded components with as much confidence for standardization as the selected subset. Excluded components remain in the Folly implementation. There is still value in receiving feedback about the excluded components of TS2.

1.5 Guidance for ABI-Stability

The following guidance covers three potential extensions to the hazard pointer interface: (1) integrated commutative counting, (2) cohort-based synchronous reclamation, and (3) custom domains. Based on the experience with Folly in production there is high confidence in the usefulness and stability of the first two items (in heavy use and ABI-stable since 2017 and 2018, respectively). The only reason they are not included in this proposal is that we expect their wording to be complicated and that the current proposal is definitely useful on its own. As for the third item, custom domains, it is unclear if it is useful or not without additional user experience and feedback.

For each extension we provide guidance for reserved space and existing functions (in particular inlined and header functions) that may be affected by the extension.

Guidance for likely future addition of integrated commutative counting:

- Reserved space: A counter (e.g., 32-bits) in `hazard_pointer_obj_base`.
- Effect on existing functions:
 - `hazard_pointer_obj_base` constructor: Initialize the counter to some fixed value (e.g., zero).

Guidance for likely future addition of cohort-based synchronous reclamation:

- Reserved space: A pointer in `hazard_pointer_obj_base`.
- Effect on existing functions:
 - `hazard_pointer_obj_base` constructor: Initialize the pointer to null.
 - `hazard_pointer_obj_base::retire`: Check the reserved pointer and if not null then call some non-header function stub.

Guidance for possible future addition of custom domains:

- Reserved space: A pointer in some internal structure associated with the actual hazard pointers. For example, in Folly, each `hazard_pointer` object contains only a single pointer to an internal hazard pointer structure which in turn contains the actual hazard pointer and a pointer to the associated domain. The rationale for not including the domain pointer in the `hazard_pointer` object itself is to minimize the latency of `hazard_pointer` object construction/destruction.
- Effect on existing functions:
 - `hazard_pointer` destructor (expected to be inlined in a header and to have very low latency): Check that the internal domain pointer points to the default domain (or alternatively use null to represent the default domain).
 - Other effects are expected to be only in non-header library functions (that do not need to be fast) that handle the creation of actual new hazard pointers (in contrast to `hazard_pointer` objects).

2 Hazard pointer overview

A hazard pointer is a single-writer multi-reader pointer that can be owned by at most one thread at any time. Only the owner of the hazard pointer can set its value, while any number of threads may read its value. A thread that is about to access dynamic objects acquires ownership of hazard pointer(s) to protect such objects from being reclaimed. The owner thread sets the value of a hazard pointer to point to an object in order to indicate to concurrent threads — that may remove such object — that the object is not yet safe to reclaim.

Hazard pointers are owned and written by threads that act as accessors/protectors (i.e., protect removable objects from unsafe reclamation in order to access such objects) and are read by threads that act as removers/reclaimers (i.e., may remove and try to reclaim objects). Removers retire removed objects to the hazard pointer library (i.e., pass the responsibility for reclaiming the objects to the library code rather than normally by user code). The set of protector and remover threads may overlap.

The key rule of the hazard pointers method is that **a retired object can be reclaimed only after it is determined that no hazard pointers have been pointing continuously to it from a time before its retirement.**

The core basic mechanism of the hazard pointer technique is the interaction between protection and deferred reclamation.

Protection:

- By setting a hazard pointer H to the address of an object A , the owner of the hazard pointer is telling all threads: “if you (collectively) remove object A after I have set H to the address of A , and then you retire A , then don’t reclaim A as long as H continues to point to A ”.

Deferred reclamation:

- After accumulating a number of retired objects (for the sake of amortization):

- Extract the set of retired objects.
- Read (no atomicity needed) the values of all hazard pointers and keep a private set of such values.
- For each retired object, lookup its address in the private set of values read from hazard pointers:
 - * If not found, reclaim the object.
 - * If found, put the object back in the set of retired objects.

The main user operations on hazard pointers are:

- Acquiring ownership of a hazard pointer.
- Setting the value of a hazard pointer to protect an object.
- Clearing the value of a hazard pointer.
- Releasing ownership of a hazard pointer.

The main user operation on objects protectable by hazard pointers in addition to regular operations such as reading, writing and removal is:

- Retiring a removed object.

The main internal library operations are:

- Allocation of hazard pointers.
- Deferred reclamation.

3 Use examples

3.1 Read-Mostly Shared Data

Without hazard pointers	With hazard pointers
<pre>struct Data { /* members */ };</pre>	<pre>struct Data : hazard_pointer_obj_base<Data> { /* members */ };</pre>
<pre>Data* pdata_; shared_mutex m_;</pre>	<pre>atomic<Data*> pdata_;</pre>
<pre>// Called frequently and in parallel template <typename U, typename Func> U reader_op(Func userFn) { shared_lock<shared_mutex> rlock(m_); // The user function userFn is not allowed // to block or call the writer function // because the read lock blocks writers. return userFn(pdata_); }</pre>	<pre>// Called frequently and in parallel template <typename U, typename Func> U reader_op(Func userFn) { hazard_pointer h = make_hazard_pointer(); Data* p = h.protect(pdata_); // The user function userFn is allowed to block // and call the writer function because hazard // pointer protection does not block writers or // prevent the reclamation of unprotected objects. return userFn(p); // Safe to access *p. } // RAII end of protection.</pre>
<pre>// May be called concurrently with readers void writer(Data* newdata) { Data* old; { unique_lock<shared_mutex> wlock(m_); old = exchange(pdata_, newdata); } delete old; // Reclaim old immediately. }</pre>	<pre>// May be called concurrently with readers void writer(Data* newdata) { Data* old = pdata_.exchange(newdata); old->retire(); // Reclaim old when unprotected. }</pre>

Typical latency of `hazard_pointer` construction/destruction in the Folly implementation on a contemporary commodity server is approximately 4 ns. Using a pre-constructed `hazard_pointer` typically takes under one nano second for protection.

3.2 Iteration

The proposed hazard pointer interface is capable of supporting iterators. For example the Folly `ConcurrentHashMap` (excluding the support of synchronous reclamation using cohorts) is fully implementable using this proposed interface.

(Detailed implementation in github.com/facebook/folly/blob/master/folly/concurrency/ConcurrentHashMap.h).

Using this proposal for hazard pointers, a container can support fast and scalable iterators that are allowed to be kept active indefinitely without blocking the progress of other functions or preventing the reclamation of unprotected objects. A user can write the following code for some shared container `x`:

```
for (auto it = x.begin(); it != x.end(); ++it) userFn(it);
```

Note that iteration can be fast (e.g., Folly `ConcurrentHashMap` can achieve 7 ns per iteration through a cache-warmed 100K-item integer to integer map). It can also be perfectly scalable for large numbers of threads to iterate in parallel on the same shared container without iterator operations interfering with each other. Also, the

user function in the example is allowed to block or have other dependencies without concern about affecting liveness or preventing reclamation of unprotected objects.

3.3 Search ordered single-writer singly-linked list

```
struct Node : hazard_pointer_obj_base<Node> {
    T elem_;
    atomic<Node*> next_;
    Node(T e, Node* n) : elem_(e), next_(n) {}
};

atomic<Node*> head_{nullptr};

bool contains(const T& val) const {
    /* Two hazard pointers for hand-over-hand traversal */
    hazard_pointer hptr_prev = make_hazard_pointer();
    hazard_pointer hptr_curr = make_hazard_pointer();
    while (true) {
        atomic<Node*>* prev = &head_;
        Node* curr = prev->load(std::memory_order_acquire);
        while (true) {
            if (!curr) return false;
            if (!hptr_curr.try_protect(curr, *prev)) break;
            Node* next = curr->next_.load(std::memory_order_acquire);
            if (prev->load(std::memory_order_acquire) != curr) break;
            if (curr->elem_ >= val) return curr->elem_ == val;
            prev = &(curr->next_);
            curr = next;
            swap(hptr_curr, hptr_prev);
        }
    }
}
```

// For more details see github.com/facebook/folly/blob/master/folly/synchronization/example/HazptrSWMRSet.h

4 Proposed Wording

[saferecl]

4.1 Hazard pointers

[saferecl.hp]

4.1.1 General

[saferecl.hp.general]

- ¹ A hazard pointer is a single-writer multi-reader pointer that can be owned by at most one thread at any time. Only the owner of the hazard pointer can set its value, while any number of threads may read its value. The owner thread sets the value of a hazard pointer to point to an object in order to indicate to concurrent threads—that may delete such an object—that the object is not yet safe to delete.
- ² A class type *T* is *hazard-protectable* if it has exactly one base class of type `hazard_pointer_obj_base<T,D>` for some *D*, that base is public and non-virtual, and it has no base classes of type `hazard_pointer_obj_base<T2,D2>` for any other combination *T2*, *D2*. An object is *hazard-protectable* if it is of hazard-protectable type.
- ³ The time span between creation and destruction of a hazard pointer *h* is partitioned into a series of *protection epochs*; in each protection epoch, *h* either is *associated with* a hazard-protectable object, or is *unassociated*. Upon creation, a hazard pointer is unassociated. Changing the association (possibly to the same object) initiates a new protection epoch and ends the preceding one.
- ⁴ An object *x* of hazard-protectable type *T* is *retired* with a deleter of type *D* when the member function `hazard_pointer_obj_base<T,D>::retire` is invoked on *x*. Any given object *x* shall be retired at most once.
- ⁵ A retired object *x* is *reclaimed* by invoking its deleter with a pointer to *x*; the behavior is undefined if that invocation exits via an exception.
- ⁶ A hazard-protectable object *x* is *possibly-reclaimable* with respect to an evaluation *A* if:
 - (6.1) — *x* is not reclaimed; and
 - (6.2) — *x* is retired in an evaluation *R* and *A* does not happen before *R*; and
 - (6.3) — for all hazard pointers *h* and for every protection epoch *E* of *h* during which *h* is associated with *x*:
 - (6.3.1) — if the beginning of *E* happens before *R*, the end of *E* strongly happens before *A*; and
 - (6.3.2) — if *E* began by an evaluation of `try_protect` with argument `src`, label its atomic load operation *L*. If there exists an atomic modification *B* on `src` such that *L* observes a modification that is modification-ordered before *B*, and *B* happens before *x* is retired, the end of *E* strongly happens before *A*.

[Note 1: In typical use, a store to `src` sequenced before retiring *x* will be such an atomic operation *B*. — end note]

[Note 2: The latter two conditions convey the informal notion that a protection epoch that began before retiring *x*, as implied either by the happens-before relation or the coherence order of some source, delays the reclamation of *x*. — end note]
- ⁷ The number of possibly-reclaimable objects has an unspecified bound.

[Note 3: The bound can be a function of the number of hazard pointers, the number of threads that retire objects, and the number of threads that use hazard pointers. — end note]

[Example 1: The following example shows how hazard pointers allow updates to be carried out in the presence of concurrent readers. The object of type `hazard_pointer` in `print_name` protects the object `*ptr` from being reclaimed by `ptr->retire` until the end of the protection epoch.

```
struct Name : public hazard_pointer_obj_base<Name> { /* details */ };
atomic<Name*> name;

// called often and in parallel!
void print_name() {
    hazard_pointer h = make_hazard_pointer();
    Name* ptr = h.protect(name); /* Protection epoch starts */
}
```



```

    /* ... safe to access *ptr ... */
} /* Protection epoch ends. */

// called rarely, but possibly concurrently with print_name
void update_name(Name* new_name) {
    Name* ptr = name.exchange(new_name);
    ptr->retire();
}

— end example]

```

4.1.2 Header <hazard_pointer> synopsis

[saferecl.hp.syn]

```

namespace std {
    // 4.1.3, class template hazard_pointer_obj_base
    template <class T, class D = default_delete<T>> class hazard_pointer_obj_base;

    // 4.1.4, class hazard_pointer
    class hazard_pointer;

    // 4.1.5, Construct non-empty hazard_pointer
    hazard_pointer make_hazard_pointer();

    // 4.1.6, Hazard pointer swap
    void swap(hazard_pointer&, hazard_pointer&) noexcept;
}

```

4.1.3 Class template hazard_pointer_obj_base

[saferecl.hp.base]

```

template <class T, class D = default_delete<T>>
class hazard_pointer_obj_base {
public:
    void retire(D d = D()) noexcept;
protected:
    hazard_pointer_obj_base() = default;
    hazard_pointer_obj_base(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base(hazard_pointer_obj_base&&) = default;
    hazard_pointer_obj_base& operator=(const hazard_pointer_obj_base&) = default;
    hazard_pointer_obj_base& operator=(hazard_pointer_obj_base&&) = default;
    ~hazard_pointer_obj_base() = default;

private:
    D deleter; // exposition only
};

```

- 1 D shall be a function object type ([func.require]) for which, given a value d of type D and a value ptr of type T*, the expression d(ptr) is valid.
- 2 The behavior of a program that adds specializations for hazard_pointer_obj_base is undefined.
- 3 D shall meet the requirements for Cpp17DefaultConstructible and Cpp17MoveAssignable.
- 4 T may be an incomplete type. It shall be complete before any member of the resulting specialization of hazard_pointer_obj_base is referenced.

```
void retire(D d = D()) noexcept;
```

- 5 *Mandates:* T is a hazard-protectable type.
- 6 *Preconditions:* *this is a base class subobject of an object x of type T. x is not retired. Move-assigning d to deleter does not exit via an exception.
- 7 *Effects:* Move-assigns d to deleter, thereby setting it as the deleter of x, then retires x. May reclaim possibly-reclaimable objects.

4.1.4 Class hazard_pointer

[saferecl.hp.holder]

4.1.4.1 Synopsis

[saferecl.hp.holder.syn]

```
class hazard_pointer {
public:
    hazard_pointer() noexcept;
    hazard_pointer(hazard_pointer&&) noexcept;
    hazard_pointer& operator=(hazard_pointer&&) noexcept;
    ~hazard_pointer();

    [[nodiscard]] bool empty() const noexcept;
    template <class T> T* protect(const atomic<T*>& src) noexcept;
    template <class T> bool try_protect(T*& ptr, const atomic<T*>& src) noexcept;
    template <class T> void reset_protection(const T* ptr) noexcept;
    void reset_protection(nullptr_t = nullptr) noexcept;
    void swap(hazard_pointer&) noexcept;
};
```

- ¹ An object of type `hazard_pointer` is either empty or *owns* a hazard pointer. Each hazard pointer is owned by exactly one object of type `hazard_pointer`.

[*Note 1*: An empty `hazard_pointer` object is different from a `hazard_pointer` object that owns an unassociated hazard pointer. An empty `hazard_pointer` object does not own any hazard pointers. — *end note*]

4.1.4.2 Constructors

[saferecl.hp.holder.ctor]

```
hazard_pointer() noexcept;
```

- ¹ *Postconditions*: `*this` is empty.

```
hazard_pointer(hazard_pointer&& other) noexcept;
```

- ² *Postconditions*: If `other` is empty, `*this` is empty. Otherwise, `*this` owns the hazard pointer originally owned by `other`; `other` is empty.

4.1.4.3 Destructor

[saferecl.hp.holder.dtor]

```
~hazard_pointer();
```

- ¹ *Effects*: If `*this` is not empty, destroys the hazard pointer owned by `*this`, thereby ending its current protection epoch.

4.1.4.4 Assignment

[saferecl.hp.holder.assign]

```
hazard_pointer& operator=(hazard_pointer&& other) noexcept;
```

- ¹ *Effects*: If `this == &other` is true, no effect. Otherwise, if `*this` is not empty, destroys the hazard pointer owned by `*this`, thereby ending its current protection epoch.

- ² *Postconditions*: If `other` was empty, `*this` is empty. Otherwise, `*this` owns the hazard pointer originally owned by `other`. If `this != &other` is true, `other` is empty.

- ³ *Returns*: `*this`.

4.1.4.5 Member functions

[saferecl.hp.holder.mem]

```
[[nodiscard]] bool empty() const noexcept;
```

- ¹ *Returns*: true if and only if `*this` is empty.

```
template <class T> T* protect(const atomic<T*>& src) noexcept;
```

- ² *Effects*: Equivalent to

```
T* ptr = src.load(memory_order::relaxed);
while (!try_protect(ptr, src)) {}
return ptr;
```

```

template <class T> bool try_protect(T*& ptr, const atomic<T*>& src) noexcept;
3   Mandates: T is a hazard-protectable type.
4   Preconditions: *this is not empty.
5   Effects: Performs the following steps in order:
(5.1)   — Initializes a variable old of type T* with the value of ptr.
(5.2)   — Evaluates reset_protection(old).
(5.3)   — Assigns the value of src.load(memory_order::acquire) to ptr.
(5.4)   — If old == ptr is false, evaluates reset_protection().
6   Returns: old == ptr.

template <class T> void reset_protection(const T* ptr) noexcept;
7   Mandates: T is a hazard-protectable type.
8   Preconditions: *this is not empty.
9   Effects: If ptr is a null pointer value, invokes reset_protection(). Otherwise, associates the hazard
pointer owned by *this with *ptr, thereby ending the current protection epoch.
10  Complexity: Constant.

void reset_protection(nullptr_t = nullptr) noexcept;
11  Preconditions: *this is not empty.
12  Postconditions: The hazard pointer owned by *this is unassociated.
13  Complexity: Constant.

void swap(hazard_pointer& other) noexcept;
14  Effects: Swaps the hazard pointer ownership of this object with that of other.
    [Note 1: The owned hazard pointers, if any, remain unchanged during the swap and continue to be associated with
    the respective objects that they were protecting before the swap, if any. No protection epochs are ended or initiated.
    — end note]
15  Complexity: Constant.

```

4.1.5 make_hazard_pointer [saferecl.hp.make]

```

hazard_pointer make_hazard_pointer();
1   Effects: Constructs a hazard pointer.
2   Returns: A hazard_pointer object that owns the newly-constructed hazard pointer.
3   Throws: May throw bad_alloc if memory for the hazard pointer could not be allocated.

```

4.1.6 hazard_pointer specialized algorithms [saferecl.hp.special]

```

void swap(hazard_pointer& a, hazard_pointer& b) noexcept;
1   Effects: Equivalent to a.swap(b).

```

4.2 Annex B (normative) Implementation quantities [implimits]

Append the following:

- Number of hazard-protectable possibly-reclaimable objects [256].

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References

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